

Epidemiology and Prognostic Impact of Acute Kidney Injury and Acute Kidney Disease

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INCIDENCE OF ACUTE KIDNEY INJURY

Epidemiologic studies have primarily identified acute kidney injury (AKI) through administrative data or changes in serum creatinine without specifying etiology. These studies have primarily focused on acute tubular necrosis in hospitalized populations and may not be generalizable to other forms of AKI. The incidence of AKI in unselected hospitalized patients is between 0.4% and 18%, depending on the definition used, and accounts for 1% to 4% of all hospital admissions.¹ Large studies using administrative data suggest that the incidence of AKI in hospitalized patients has increased by approximately 13% per year over the last 3 decades.² In these studies, incidence was identified by diagnostic codes that are highly specific for AKI (97%) but are relatively insensitive (35% sensitivity).³ The findings of such studies may be influenced by increasing clinical recognition of less severe AKI, leading to increased coding of AKI without a change in true incidence. In addition, there may be financial incentives to code patients as having AKI to increase reimbursement. However, similar increases have been observed in the incidence of AKI requiring kidney replacement therapy (KRT): in general, populations of hospitalized patients, as well as in specific settings including surgery, sepsis, stroke, cardiovascular events, and liver disease.⁴ Although more objective and clinically relevant than diagnostic codes, dialysis utilization may also be influenced by increased availability of KRT or changing trends in medical practice rather than true changes in the burden of disease.

The incidence of AKI can be estimated using consensus definitions based on serum creatinine and urine output (Table 73.1) and has been best characterized in critically ill populations. Despite using standardized definitions for AKI, multicenter studies of critically ill patients have reported the incidence of AKI to be between 10% and 67%, likely reflecting differences in case mix between health care systems and countries.⁵

Less is known about AKI in the community; there is significant etiology and geographic variation in the reported incidence and risk factors for AKI in outpatient settings, where serum creatinine is infrequently measured. The concept of acute kidney disease (AKD) was proposed in the 2012 Kidney Disease: Improving Global Outcomes guidelines in recognition that subacute kidney injury may not meet criteria for AKI or chronic kidney disease (CKD) despite being clinically relevant (Fig. 73.1). AKI is recognized as a subset of AKD; the relationships between AKI, AKD, CKD, and the absence of kidney disease are shown in Fig. 73.2. A population-based study in Alberta, Canada ($n = 921,116$) compared the incidence and prognosis of AKD compared with AKI and CKD. AKD was 2.7-fold more prevalent than AKI, affecting 4.4% of the cohort, and was primarily identified in outpatients (77%).⁶

RISK FACTORS FOR AKI

Risk factors for AKI have been determined in a variety of clinical settings including cardiac surgery, contrast-induced AKI, and critically ill

populations. However, risk prediction models that accurately predict the occurrence of AKI have been elusive. Nonmodifiable risk factors common to all populations are presented in Box 73.1.

Age

Multiple studies show that AKI is more common in older individuals, and many have shown an independent association between AKI and older age.⁷ In a community-based prospective study, patients aged 80 to 89 years were 55 times more likely to develop AKI than adults younger than 50 years.⁸ Possible explanations for this association include (1) structural and functional changes associated with age that lead to diminished nephron reserve and reduced capacity for kidney autoregulation, (2) accumulation of comorbidity that increases susceptibility to AKI (vascular disease, diabetes, hypertension, CKD), and (3) increased exposure among older persons to medications and procedures that predispose to AKI.⁷

Reduced Estimated Glomerular Filtration Rate

Preexisting reduction in estimated glomerular filtration rate (eGFR) is a potent risk factor for AKI after exposure to radiocontrast,⁹ major surgery, and medical illness,¹⁰ although the pathophysiology underlying this association is poorly understood. It has been reported that the odds of developing dialysis requiring AKI are increased at lower baseline eGFR; the excess risk compared with normal eGFR was approximately twofold in patients with baseline eGFR 45 to 60 mL/min/1.73 m², but more than 40-fold for patients with baseline eGFR less than 15 mL/min/1.73 m².¹¹ These associations were confirmed in several systematic reviews that demonstrate strong independent associations between risk for AKI and lower baseline eGFR¹²⁻¹⁴ (see Fig. 73.3). Although these analyses support a causal association between CKD and in-hospital AKI, little is known about how this association may be modified by the presence of one or more comorbidities, such as heart failure, or whether all causes of CKD confer similar risk for AKI.

Albuminuria

Albuminuria is also strongly associated with AKI risk. A case-control study of over 600,000 patients identified proteinuria as an independent predictor for AKI,¹¹ which was replicated in multiple settings, including after cardiac surgery in Taiwan and in general population studies from the United States and Canada¹⁵⁻¹⁷ (Fig. 73.4).

Hyperuricemia

Two prospective randomized trials^{18,19} also found allopurinol plus hydration to be superior to hydration alone in preventing AKI following radiocontrast administration. Nevertheless, uric acid can also be a biomarker for other risk factors (e.g., mild CKD, cancer), and although the two studies are interesting, more data is needed before this intervention is recommended. A recent systematic review and meta-analysis

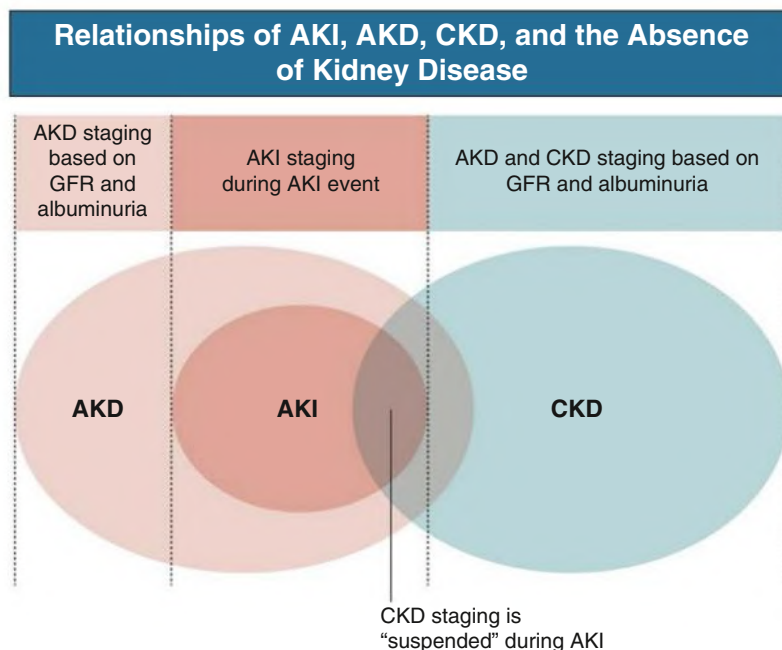


Fig. 73.2 Relationships of AKI, AKD, CKD, and the absence of kidney disease according to KDIGO definitions. AKD, Acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; GFR, glomerular filtration rate; KDIGO, Kidney Disease: Improving Global Outcomes; NKD, no kidney disease; SCr, serum creatinine. (From Lameire NH, Levin A, Kellum JA et al. Harmonizing acute and chronic kidney disease definition and classification: report of a Kidney Disease: Improving Global Outcomes [KDIGO] Consensus Conference. *Kidney Int.* 2012;100[3]:516–526.)

BOX 73.1 Patient-Specific Risk Factors for Acute Kidney Injury

- Age
- Sex (male)
- Chronic kidney disease
- Proteinuria
- Diabetes
- Congestive heart failure
- Sepsis
- Volume depletion
- Chronic liver disease
- Hyperuricemia

AKI. Therefore, there is uncertainty about whether the association between AKI and mortality is truly independent. Nonetheless, the development of AKI signifies a poor prognosis in individual patients.

Small changes in serum creatinine of as little as 25% above baseline are a significant predictor of all-cause short-term mortality.²⁴ A meta-analysis of eight studies of hospitalized patients (most of whom were critically ill or had heart failure) confirmed a graded relationship between increasing severity of AKI and short-term mortality.²⁵ Most importantly, it confirmed that even mild forms of AKI are clinically relevant; an increase in serum creatinine of 0.3 mg/dL (26 μ mol/L) was associated with a 2.3-fold increase in the relative risk for death. After adjustment for comorbidities, the ORs or hazard ratios (HRs) for death associated with AKI ranged from 1.7 to 1.92. Similar findings were reported in several population-based studies of unselected hospitalized patients.²⁶

The association between AKI and mortality is likely influenced by several factors, including the presence of underlying CKD, the duration

and severity of AKI, and the degree of recovery of kidney function. A population-based study of hospitalized AKI patients in Canada found that the relation between AKI and in-hospital mortality was strongest in those with severe AKI and baseline eGFR greater than 60 mL/min/1.73 m², for whom the adjusted HR suggested a more than tenfold increase in the risk of death compared with those without AKI.^{26,27} The presence of CKD modifies this association; by comparison, in patients with a baseline eGFR less than 30 mL/min/1.73 m², the adjusted HR of in-hospital mortality was increased fivefold compared with those without CKD or AKI. An analysis of postoperative AKI comparing kidney and survival outcomes in those with and without preexisting CKD (eGFR <45 mL/min/1.73 m²) found a lower attributable mortality rate because of AKI (HR 1.26, 95% CI 1.09–1.78) when subjects with prior CKD but no AKI were used as the reference.²⁸

Although the incidence of AKI continues to climb, there has been a corresponding improvement in survival. A recent analysis reported a 19% decrease in mortality between 2000 and 2009 in patients who required acute dialysis. Whether this represents a trend toward earlier use of dialysis (rather than a true improvement in survival) requires further investigation.

Chronic Kidney Disease

Until recently, it was accepted that AKI survivors generally recovered kidney function and remained independent from dialysis. This assumption was supported by the findings of small cohort studies of predominantly critically ill patients. One recent study described long-term outcomes of 226 critically ill patients with previously normal eGFR who required acute dialysis for AKI and reported that 86% of survivors had “normal” kidney function at 5 years.²⁹

However, an increasing number of recent studies have linked AKI survivorship to the development of CKD or end-stage kidney disease (ESKD). For example, in one study AKI (defined using ICD-9 codes)

Risk of Acute Kidney Injury as a Function of Baseline Kidney Function (Albuminuria and eGFR) for People With and Without Diabetes

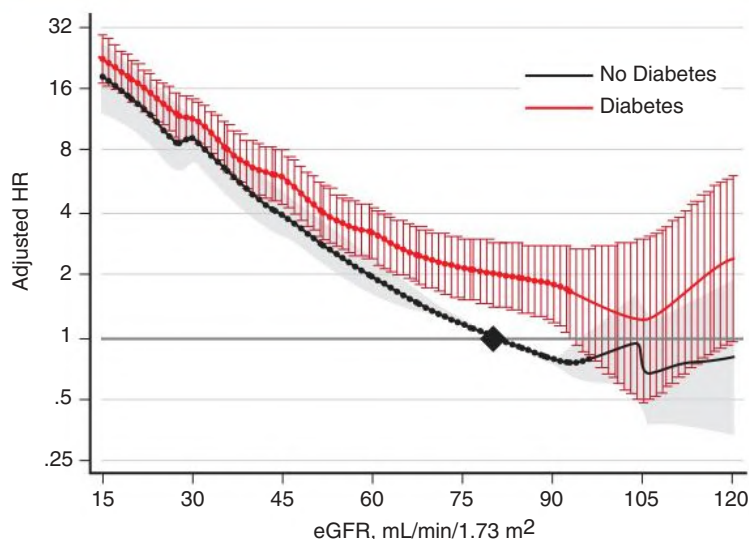


Fig. 73.3 Hazard ratio (HR) of acute kidney injury for people with and without diabetes according to baseline estimated glomerular filtration rate. *eGFR*, Estimated glomerular filtration rate.

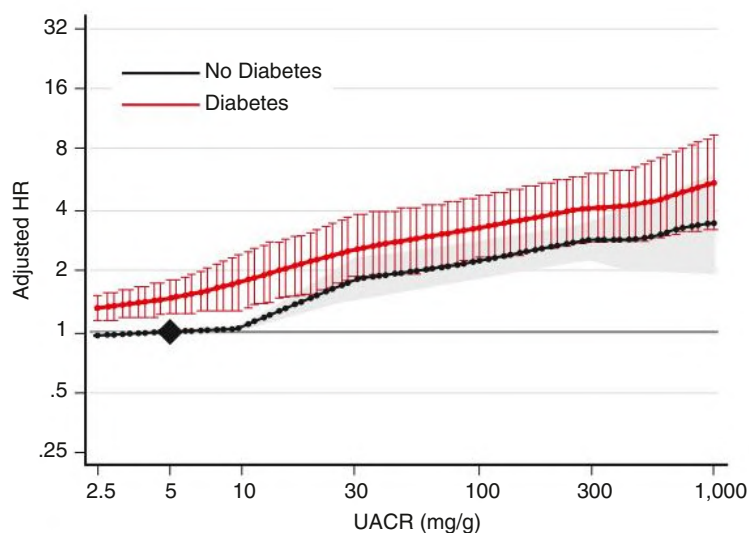


Fig. 73.4 Hazard ratio (HR) of acute kidney injury for people with and without diabetes according to the ratio of urine albumin to creatinine. *UACR*, Urine albumin/creatinine ratio.

was associated with an eightfold increase in the risk for ESKD compared with subjects without AKI or CKD.³⁰ A second cohort study evaluated the risk for progressive CKD after AKI in subjects with baseline eGFR greater than 45 mL/min/1.73 m², finding that survivors of dialysis-dependent AKI had a 28-fold increased risk for developing stage 4 CKD (eGFR <30 mL/min/1.73 m²) or greater, compared with those without AKI.³¹

Since then, multiple studies have demonstrated an association between AKI and initiation or progression of CKD and/or ESKD in a variety of clinical settings, including cardiac surgery, cardiac catheterization, and in unselected hospitalized patients. A meta-analysis of 13 cohort studies reported that the HRs for CKD and ESKD were 8.8 (95% CI 3.1–25.5) and 3.1 (95% CI 1.9–5.0), respectively, compared with subjects without AKI. Kidney recovery after AKI appears to be

an important determinant of future kidney function;³² a Canadian population-based study showed that AKI that failed to return to within 25% of baseline kidney function was associated with a fourfold risk of CKD/ESKD.³³ National kidney registry data from the United States shows that approximately 2% to 3% of patients with AKI requiring dialysis will fail to recover kidney function and/or progress to ESKD within 1 year.³⁴ Recurrent episodes of AKI further increase the risk for progressive CKD; each additional AKI event after the first episode appears to double the risk for progression to stage 4 CKD.

It is possible that the apparent association between AKI and CKD is confounded by age, frailty, and unmeasured comorbidity. Also, it is sometimes difficult to distinguish AKI from progressive CKD, and there is the possibility that the observed associations are due to misclassification. However, AKI also has been linked to the development

BOX 73.2 Risk Factors for Chronic Kidney Disease After Acute Kidney Injury

- Age
- Baseline eGFR
- Congestive heart failure
- Hypertension
- Recurrent AKI
- Serum albumin during hospitalization
- Severity of AKI (AKI stage, requirement for dialysis)
- eGFR at hospital discharge

AKI, Acute kidney injury; eGFR, estimated glomerular filtration rate.

of CKD in children, in whom these potential confounders are less likely.³⁵ A prospective single-center study of 126 critically ill pediatric AKI survivors who fully recovered kidney function reported that 10% of patients developed CKD (albumin-to-creatinine ratio ≥ 30 mg/g or eGFR < 60 mL/min/1.73 m²) over 3 years of follow-up.³⁶ Importantly, 38% of the cohort had mildly decreased eGFR (60–90 mL/min/1.73 m²), 3.2% had hypertension, and 8.7% had hyperfiltration—all risk factors for future CKD. Given that the median age of these patients at the time of the AKI episode was 0.5 years, these numbers likely underestimate the long-term risk for CKD for these children. Similar findings were reported in a meta-analysis of outcomes after pediatric hemolytic uremic syndrome.³⁷

Cardiovascular Risk

Contrast-induced nephropathy after cardiac catheterization is known to be associated with excess risk for subsequent cardiovascular events; however, there are conflicting data about whether AKI is associated with an increased risk for cardiovascular events in patients without preexisting cardiovascular disease. Several large retrospective cohort studies of patients undergoing both vascular and nonvascular major surgery have recently confirmed an association between postoperative AKI and cardiovascular mortality.^{38,39} Further, a recent systematic review analyzed data from 25 cohort studies of patients ($n = 254,408$) with and without AKI and reported that AKI is associated with an 86% increased risk for cardiovascular mortality, a 38% increased risk for major adverse cardiovascular events, and a 40% increased risk for heart failure.⁴⁰ The pathophysiology of this association remains unclear.

In summary, the development of AKI during hospitalization identifies a cohort of patients at high risk for adverse outcomes. Recognition of AKI therefore presents an opportunity to mitigate

future risk through appropriate monitoring and implementation of evidence-based care for CKD. However, US data suggest that only 5% of AKI survivors saw a nephrologist after hospital discharge,³⁴ and AKI survivors with CKD and proteinuria often do not receive treatment with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers after discharge from hospital.⁴¹ This treatment gap is undoubtedly larger in patients with AKI from lower-income countries.

Health Care Costs

A single-center study of AKI in hospitalized patients demonstrated a direct relationship between severity of AKI and associated hospital length of stay and hospital costs.²⁴ When defined as a 0.3 mg/dL (26 μ mol/L) increase in serum creatinine, AKI was associated with an incremental hospitalization cost of \$4886; a doubling of serum creatinine was associated with an incremental cost of \$9000. Studies of specific populations of hospitalized patients support these findings; a recent study of the cost of AKI after cardiac surgery suggests that the average difference in postoperative costs ranges between \$9000 and \$14,000 depending on AKI severity. Postoperative AKI in noncardiac surgery is associated with an \$11,308 increase in the median cost compared with patients who do not experience postoperative AKI.⁴² In multivariable analysis, AKI was the most costly postoperative complication for these patients and resulted in the largest proportion of resource use compared with all other complications. Another study of 5875 surgical patients found severe AKI requiring KRT to be the second most costly postoperative complication, with an estimated mean increase in hospital expenditures of \$28,359 compared with an uncomplicated postoperative course, and resulted in almost twice the excess cost compared with cardiac arrest.⁴³ However, none of these studies accounted for the impact of CKD on AKI and attendant costs, which is likely to be significant.

AKI AS A PUBLIC HEALTH ISSUE

Conservative population-based estimates of AKI incidence in hospitalized adults are in the range of 3000 per 100,000 person-years;⁴⁴ the majority of these patients will survive to hospital discharge. The incidence of stage 4 CKD or greater in AKI survivors is approximately 120 per 100,000 person-years,⁴⁵ and the number of adults developing new CKD after AKI may be as high as 100,000 per year in the United States alone. Extrapolated to a global scale, AKI is likely responsible for hundreds of thousands of new CKD cases each year. Given this tremendous burden of morbidity and mortality, strategies that mitigate the progression of CKD after AKI should be a public health priority.

SELF-ASSESSMENT QUESTIONS

- All of the following are risk factors for AKI except:
 - age.
 - CKD.
 - congestive heart failure.
 - thyroid disorders.
 - diabetes.
- The incidence of AKI in hospitalized patients has:
 - decreased by more than 10% annually over the past decade.
 - decreased by 1% over the past decade.
 - increased by more than 10% annually over the past decade.
 - increased by 1% annually over the past decade.
 - stayed the same.
- AKI is associated with:
 - higher health care costs.
 - increased risk for CKD.
 - increased risk for chronic dialysis.
 - all the above.
- The incidence of stage 4 CKD (eGFR < 30 mL/min/1.73 m²) or greater in AKI survivors is approximately:
 - 1200 per 100,000 person-years.
 - 120 per 100,000 person-years.
 - 12 per 100,000 person-years.
 - 1.2 per 100,000 person-years.

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